Measuring what users actually say

A validation-first NLP study of 110,000 reviews of five Bangladeshi mobile finance apps

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Group academic project (IBA, 2025 to 2026). I drove the pipeline development and the analysis. October 2026.

Between May 2018 and August 2026, bKash, Nagad, Rocket, Upay and TAP collected about 110,700 text reviews on Google Play. A first pass with a sentiment model put 29.8% of them in the negative class. After post-stratification weighting the figure is 17.0%. The same pipeline showed that its two classifiers disagree on 57% to 64% of reviews written in Bengali or Banglish (Bengali in Latin letters), the languages a large share of local users write in. This summary covers what I built, what the data show, and how I checked the results. The checking took more of the effort than the modelling, and I think it is the more useful part.

110,721 text reviews analysed, 5 apps, 2018–2026	**29.8% → 17.0%** negative share, raw vs. weighted	**24% vs. 57–64%** model disagreement: English vs. Bengali / Banglish	**138** app releases evaluated with difference-in-differences

1. What I built

The pipeline scrapes reviews, detects the script each review is written in, classifies sentiment with a cascade of two independent models, and reweights the results to the app population using post-stratification. On top of that sit aspect tagging (eight topics), a churn-intent flag, near-duplicate detection with MinHash (Broder, 1997), reply analysis by severity tier, and release-impact estimation. Every step that produces a number has a matching check against human-labelled samples or against another model, shown in the lower row.

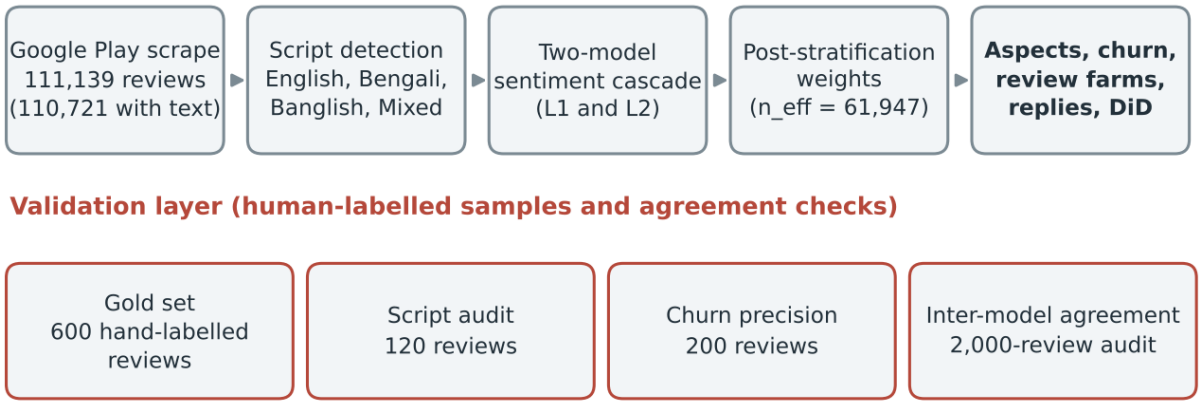


Figure 1. Pipeline and validation layer. Python; outputs delivered as a 21-page APA report and a 33-slide deck.

2. The headline number depends on weighting and on one threshold

The reviews are not a clean random sample of users, so unweighted shares are biased. Post-stratification brings the effective sample size to 61,947 (design effect 1.79; Kish, 1965) and shifts the negative share from 29.8% to 17.0%, with 95% bootstrap intervals (Efron & Tibshirani, 1993) about ±0.3 points wide. Those intervals describe sampling error only. They say nothing about classifier error, which is the subject of the next section.

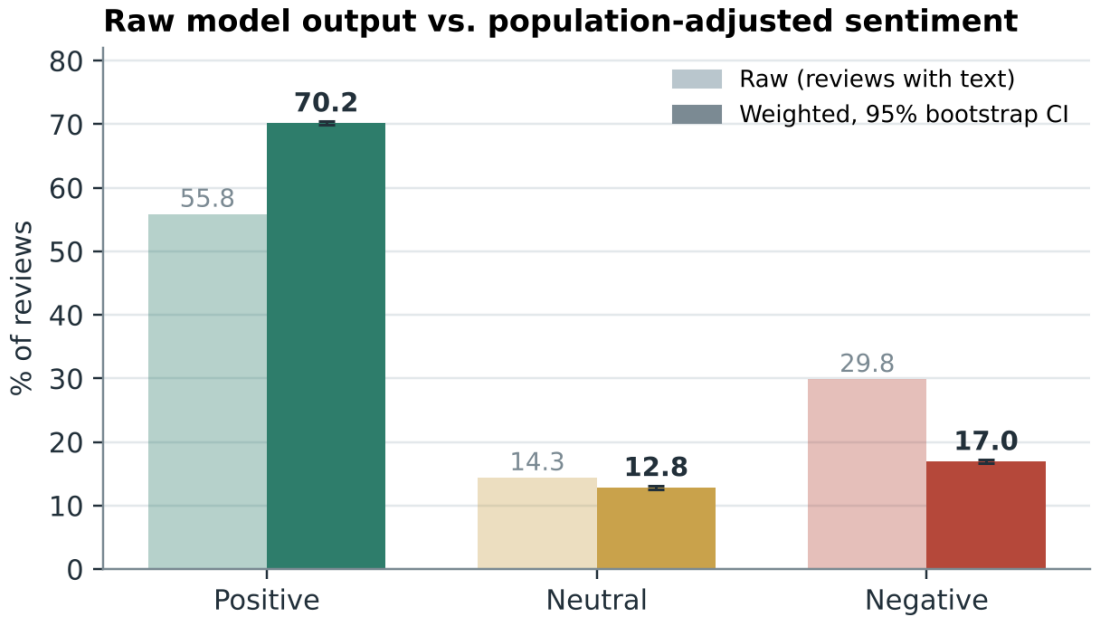


Figure 2. Raw and weighted sentiment shares, n = 110,721.

A second choice matters just as much: how wide the neutral band on the polarity score is. I swept it from 0 to 0.30 instead of fixing it. The raw negative share runs from 32.1% down to 24.5%, the neutral share from 5.5% up to 29.7%, and the positive share from 62.3% down to 45.8%. A single sentiment percentage from this kind of model is only meaningful together with its threshold.

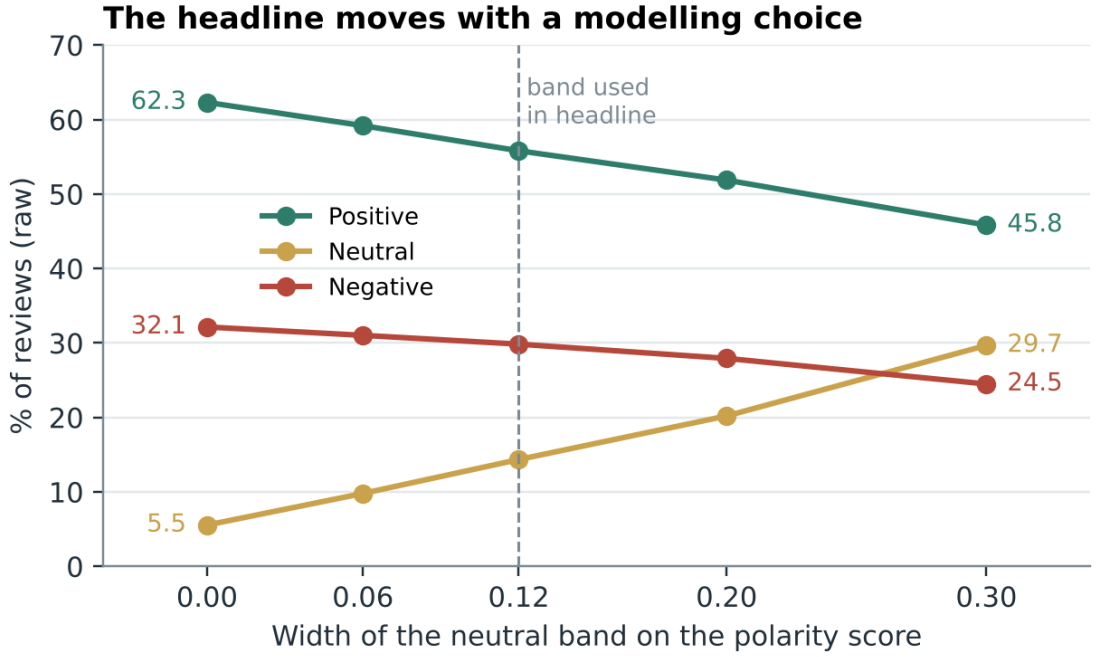


Figure 3. Sensitivity of the raw sentiment split to the neutral band. The headline used a band of 0.12.

3. Where the measurement breaks down

The two classifiers are different architectures, so their disagreement is a label-free measure of how hard a text is. On English reviews they disagree 24.1% of the time. On Bengali it is 57.2%, on Banglish 63.8%, and on mixed-script text 54.7%. Mean model confidence drops from 0.69 in English to about 0.49 in Bengali and Banglish. On a 2,000-review audit the agreement is moderate by the usual benchmarks (Landis & Koch, 1977): Cohen's κ = 0.44 (Cohen, 1960), Fleiss' κ = 0.52 across three raters, Krippendorff's α = 0.52 nominal. The ordinal α of 0.68 is higher only because it forgives near-misses between neighbouring classes, so I did not use it as the headline agreement figure.

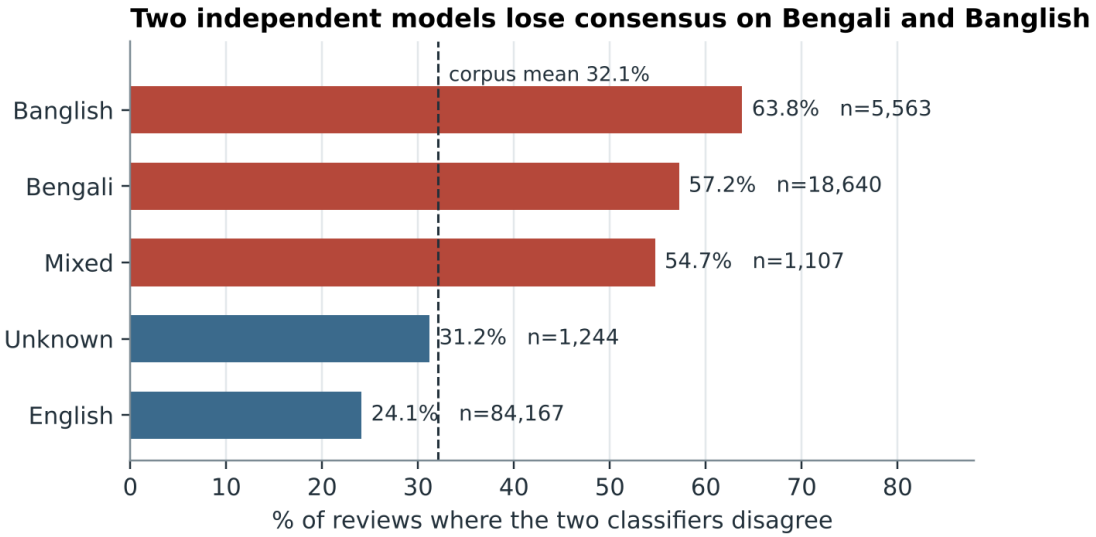


Figure 4. Share of reviews on which the two sentiment models disagree, by script (n per script shown).

Human-labelled checks (from the project report, hand-labelled samples)

Check	Result	What it means
Sentiment vs. 600-review gold set	66.5% accurate (95% CI 62.6–70.2); κ = 0.50; macro-F1 = 0.657	Accuracy falls from 80.7% on English to 58.7% on mixed text. 25.4% of genuine negatives were called neutral.
Script detector vs. 120-review audit	93.3% (112 of 120)	Weakest on Banglish (86.7%).
Churn-intent flag vs. 200-review audit	Precision 15.5% (31 of 200; Wilson 95% CI 11.1–21.2)	The raw flag overstates true churn intent about sixfold. Raw app-level rates of 1.7% to 4.5% should not be quoted without this correction.

Wilson interval: Wilson (1927).

4. What users complain about, and what the apps do about it

Performance is the clear pain point: 62% of reviews that mention it are negative (95% CI 60.6–63.7, 9,302 mentions), with an average rating of 2.26 stars. Security follows at 54% (average rating 2.20). UI/UX is the only aspect users mostly like, at 14% negative and 3.55 stars. Aspect negatives inherit the sentiment model's error, which is highest in Bengali and Banglish text, so the ordering of aspects is better supported than the exact percentages.

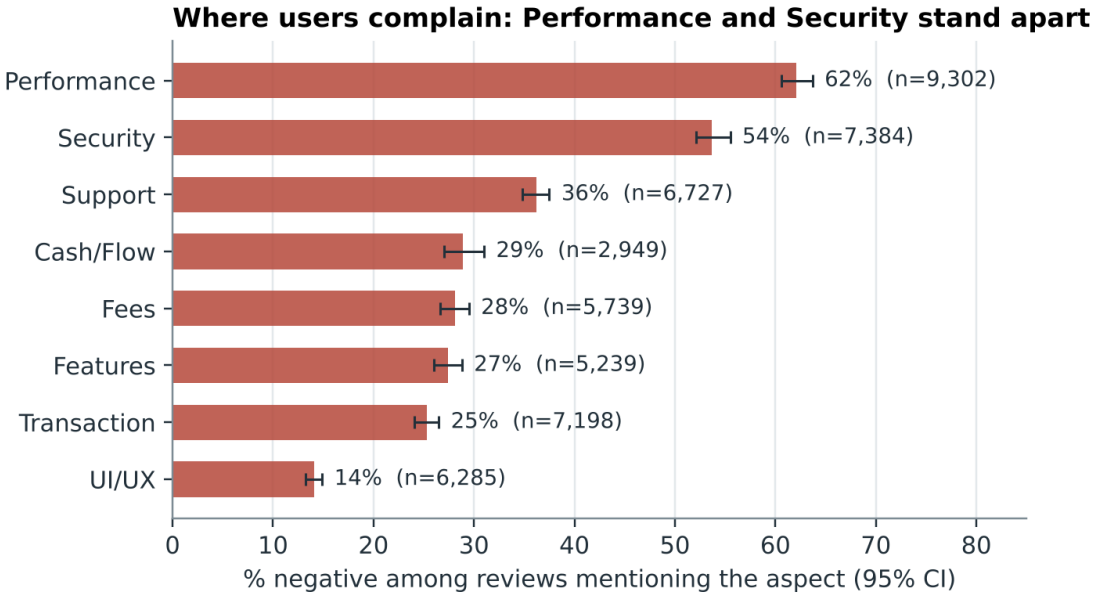


Figure 5. Negative share by aspect with 95% intervals. Mentions in parentheses.

Developer replies differ more between apps than the sentiment does. Upay replies to 99.6% of reviews and bKash to 97.4%; Nagad replies to 41.9% and Rocket to 0.9%. Inside bKash, reviews describing a financial loss get a reply 65.9% of the time, against 97.9% for reviews with no issue, so the most serious complaints are the least likely to be answered. I leave out reply latency: the timestamp data produce negative and year-long medians that I could not trust.

5. Do app releases change sentiment?

For each of 138 releases I compared the change in net sentiment around the release date with the change in the other apps over the same window, a difference-in-differences design in the spirit of Card and Krueger (1994). The median release is followed by a gain of 8.5 points relative to controls (mean +9.8), and 33% of releases show a loss. The medians by app are Upay +19.0 (11 releases), bKash +9.3 (55), Nagad +8.4 (51) and Rocket +0.5 (21). The spread is wide, and it shrinks as the review windows grow: for the 16 releases with at least 300 reviews on both sides the median is +2.5 and 44% are negative. I read the pattern as descriptive. There are no per-release intervals, and competitors ship their own updates.

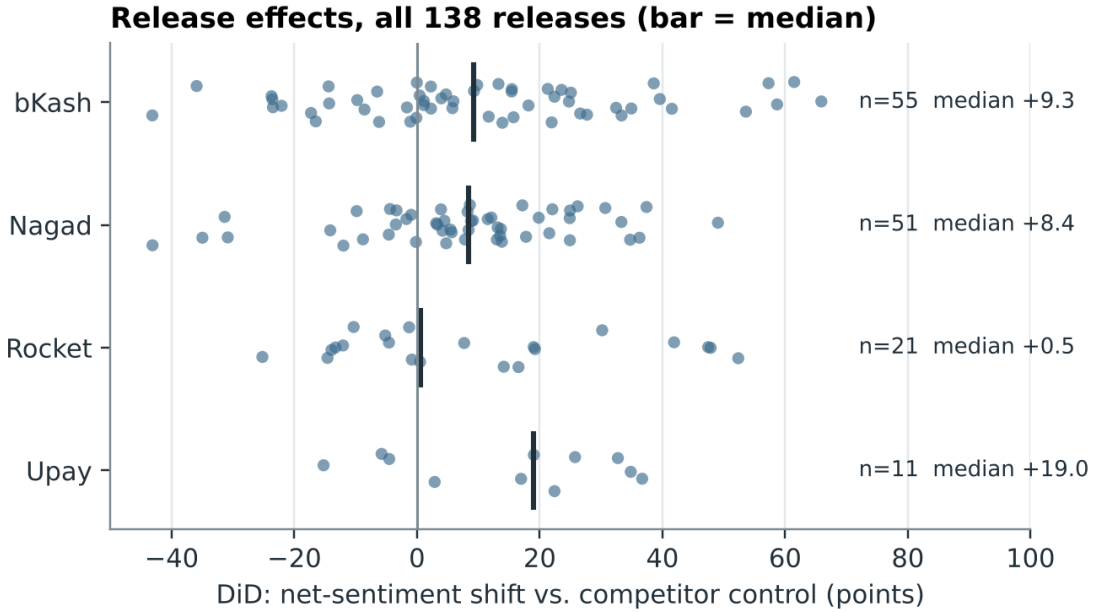


Figure 6. Difference-in-differences estimate for every release, by app. Points are releases; the bar marks the median.

6. Open issues I kept visible

- The answer-key IDs for the gold set did not line up with the annotation sheet, so I did not back-fill any labels from the key. Accuracy is reported only from labels a human entered.
- The dashboard's cross-app values and the weighted app table disagree. I flagged this in the report and have not resolved it.
- TAP has 268 reviews. It appears in the data but I draw no app-level conclusions about it.
- A very sharp break in the review series in September and October 2021 looks like a change in volume or collection, not in opinion, so I did not interpret it.

7. What the findings imply

For anyone using these apps' reviews to decide something, three things follow. Raw sentiment overstates negativity here, by roughly 13 points. Language matters: the models are weakest on the text many local users write in, so any dashboard built on them needs a language-specific check. And a regex-style churn alert would be wrong about five times in six. For the operators, the problems users name most are Performance and Security, and unanswered financial-loss complaints are where reply practice is thinnest.

8. What this says about how I would approach doctoral research

The domain here is consumer text, not thermofluids, and I am not claiming otherwise. What carries over is a way of working that applies to simulation and experiment as much as to reviews.

- **A model output is a measurement with an error.** I compared two independent models and a hand-labelled reference set. In computational work the same roles are played by code-to-code comparison and experimental data.
- **Free parameters are swept, not fixed quietly.** The neutral band here plays the part a mesh size, a turbulence model or a threshold plays in a simulation.
- **Uncertainty comes with its limits stated.** I report intervals and say which sources of error they leave out.
- **Causal claims need a design, and they weaken when the data do.** The release analysis uses controls, and I report how the estimate changes with sample size.
- **Failures stay in the document.** The unresolved issues above are in the report, not removed from it.

The practical toolkit is Python, bootstrap and agreement statistics, effective sample size, difference-in-differences, and large-scale cleaning and deduplication. I am preparing funded PhD applications in mechanical engineering (CFD, thermal sciences and sustainable energy) for Fall 2027, and I would bring this validation habit to that work.

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